

docs-pitch-one-pager

marrow — one-pager

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The embeddable agent runtime for AI that runs where you can't ship Python. Reference implementation of **ARI**, the open Agent Runtime Interface.

Problem. AI agents are moving into production (57% of teams run agents in prod — *LangChain, 2025*) and escaping the data center — into robots, edge devices, and native apps. Every mature framework (LangChain, LangGraph, CrewAI, AutoGen) is **Python-locked**, and latency is already the **#2 production blocker** (*LangChain, 2025*). You cannot ship a CPython runtime into a real-time robot or a customer's native binary.

Solution. **marrow** is a native **C++17** agent runtime — state, providers, tools, routing, lifecycle — embeddable with **no managed runtime required**, ergonomic from Python. It is the **reference implementation of ARI**, an open standard for the agent *runtime* layer that sits **beneath MCP and A2A** (complementary, not competitive).

Why it wins (the moat). ARI's **Embedded profile** is defined by constraints a Python framework *structurally cannot meet* — native core, bounded memory, mandatory cancellation/timeouts, near-zero dependencies. We own both the **standard** and its **reference implementation**, plus a **conformance kit** that makes "ARI-conformant" falsifiable by anyone. Standard + implementation + proof.

Market (cited, framed honestly). AI-agents market ~7B in 2025 at 44–50% (*syndicated; range*); humanoid robotics TAM * 38B by 2035* (*Goldman Sachs*); enterprise agentic-AI adoption **33% of apps by 2028** (*Gartner*). Our wedge (embedded/embodyed) is the fastest-growing enabling layer — an early-mover window.

Status (honest). Pre-alpha (0.1.0). Shipped: ARI spec v0.1, reference implementation, conformance kit, STRIDE threat model + security hardening + CI scanning. **External users: 0.** First deployment target: Tilo (humanoid).

Traction / Team / Ask. *Template — replace with real figures, do not present as fact:* design partners [e.g., 1 robotics + 1 trading]; founder [name + background]; raising [\$1.5M pre-seed, illustrative] for [embedded runtime, Tilo integration, a second conformant implementation, standards evangelism].

Links. Spec: ARI-SPEC.md · Strategy: docs/ari-strategy.md · Threat model: THREAT-MODEL.md · Repo: github.com/bencrooks-dev/marrow